



UITS
**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

Lab Report

Course Title: Microprocessors and Microcontrollers Lab

Course Code: CSE 360

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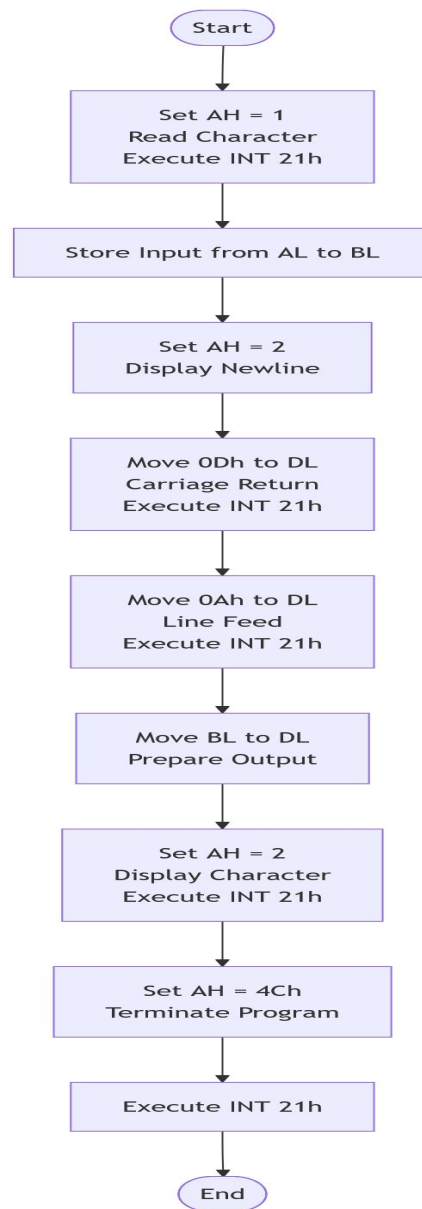
Experiment No: 01

Experiment Name:

Write an Assembly Language program to take input from the user and display the output.

Introduction:

Assembly language is a low-level programming language that provides a direct interface with the computer's hardware and microprocessor. In the 8086 architecture, interaction with the user (Input/Output) is typically handled via DOS Interrupts. Specifically, **INT 21h** is a powerful service routine used for various system tasks. In this experiment, function **01h** is used to capture a single character from the keyboard, which the processor automatically stores in the **AL** register. To display a character, function **02h** is used, which reads the data from the **DL** register and prints it to the console. This experiment also demonstrates the use of ASCII control characters like Carriage Return (**0Dh**) and Line Feed (**0Ah**) to format the output by moving the cursor to a new line.

Flowchart:**Code:**

; an Assembly Language program to take input from the user and display the output

```
ORG 100h
```

```
.code
```

```
main proc
```

```
MOV AH, 1
INT 21h
MOV BL, AL
```

```
MOV AH, 2
MOV DL, 0Dh
INT 21h
```

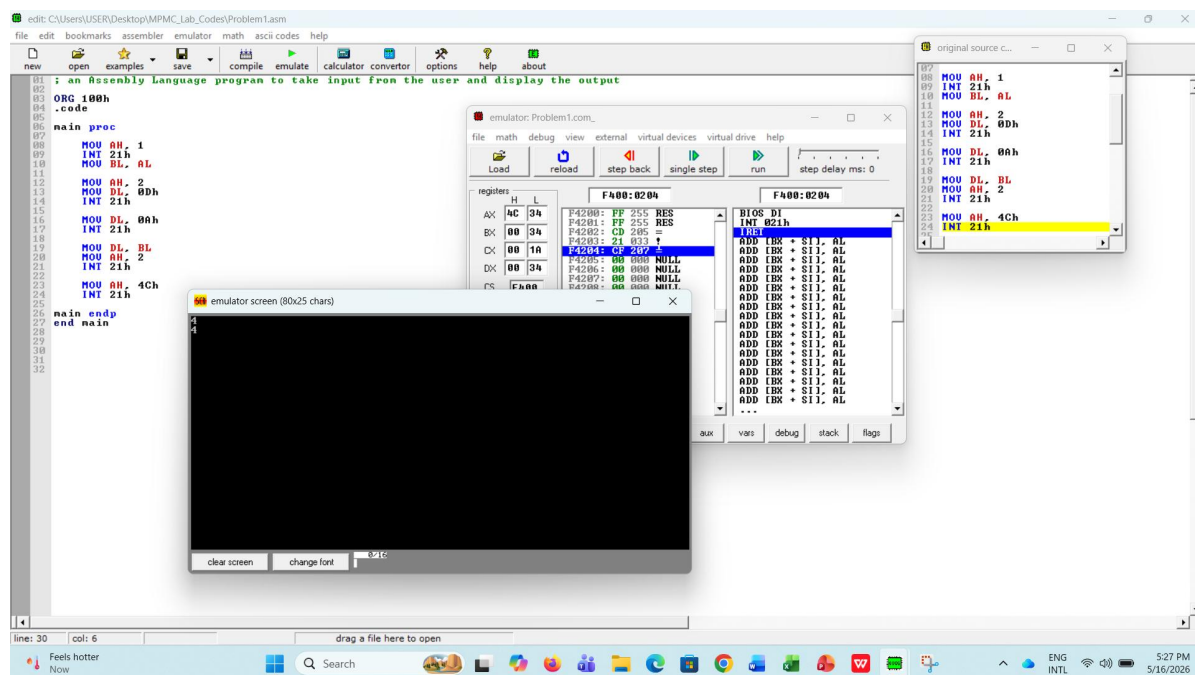
```
MOV DL, 0Ah
INT 21h
```

```
MOV DL, BL
MOV AH, 2
INT 21h
```

```
MOV AH, 4Ch
INT 21h
```

```
main endp
end main
```

Output:



Conclusion:

Through this experiment, we successfully implemented a program to capture and display user input using the 8086 microprocessor's instruction set. We gained practical experience with the INT 21h service functions and understood the necessity of moving data between registers (specially from AL to BL and then to DL) to prevent data loss during multiple interrupt calls. Additionally, the implementation of a newline using ASCII codes clarified how low-level console formatting is achieved without high-level library functions.

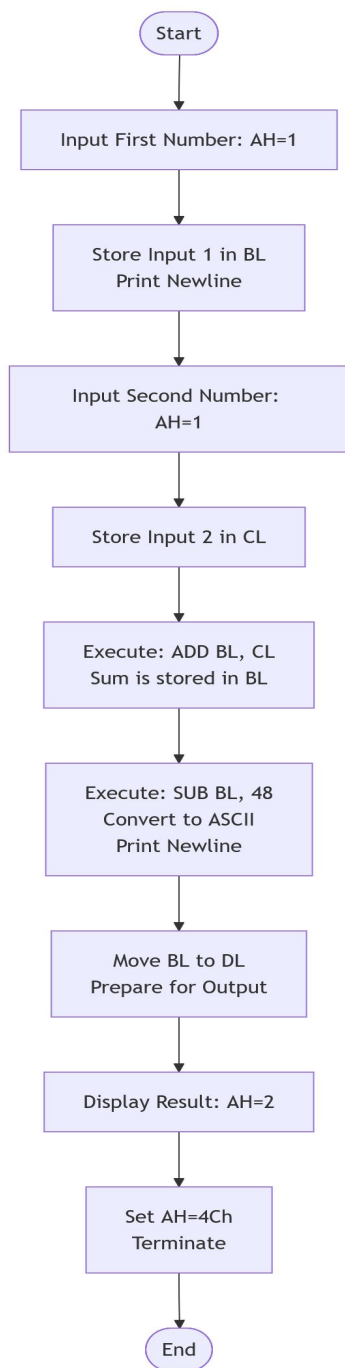
Experiment No: 02

Experiment Name:

Write an Assembly Language program to perform addition of two numbers.

Introduction:

Performing arithmetic operations is a primary function of the 8086 microprocessor. This experiment demonstrates the implementation of addition for two single-digit numbers provided by the user. When two ASCII characters are added together, the resulting sum exceeds the intended numerical value. To correct this, the program subtracts **48 (or 30h)** from the total to convert the result back into a displayable ASCII character. This experiment utilizes INT 21h function 01h for input and function 02h for output, along with the **ADD** and **SUB** instructions to manipulate the data.

Flowchart:

Code:

; an Assembly Language program to perform addition of two numbers

ORG 100h

.code

main proc

;first number

MOV AH, 1

INT 21h

MOV BL, AL

;Print Newline

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;second number

MOV AH, 1

INT 21h

MOV CL, AL

;Addition

ADD BL, CL

SUB BL, 48

;Print Newline

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;result

MOV DL, BL

MOV AH, 2

INT 21h

MOV AH, 4Ch

INT 21h

main endp

end main

Output:

The screenshot shows an 8086 assembly language emulator. The main window displays the assembly code for a program that adds two numbers. The code is as follows:

```

; an Assembly Language program to perform addition of two numbers
ORG 100h
.code
main proc
    ;first number
    MOV AH, 1
    INT 21h
    MOV BL, AL

    ;Print Newline
    MOV AH, 2
    MOV DL, 0Dh
    INT 21h
    MOV DL, 0Ah
    INT 21h

    ;second number
    MOV AH, 1
    INT 21h
    MOV CL, AL

    ;Addition
    ADD BL, CL
    SUB BL, 48

    ;Print Newline
    MOV AH, 2
    MOV DL, 0Dh
    INT 21h
    MOV DL, 0Ah
    INT 21h

    ;result
    MOV DL, BL
    MOV AH, 2
    INT 21h

    MOV AH, 4Ch
    INT 21h

main endp
end main

```

The emulator screen shows the output of the program, which is the sum of the two numbers (2 + 2 = 4) followed by a newline character. The registers window shows the current state of the CPU registers, including the AX, BX, CX, DX, SI, DI, BP, and SP registers. The status bar at the bottom shows the current line and column, and the system tray displays the date and time.

Conclusion:

In this experiment, the addition of two digits using the 8086 instruction set is performed successfully. A key observation was the behaviour of ASCII values during arithmetic; we learned that adding two ASCII-encoded digits requires a correction step (subtracting 48) to ensure the final output is numerically correct and readable on the console. This foundational logic is essential for building more complex calculators and data processing routines in low-level programming.

Experiment No: 03

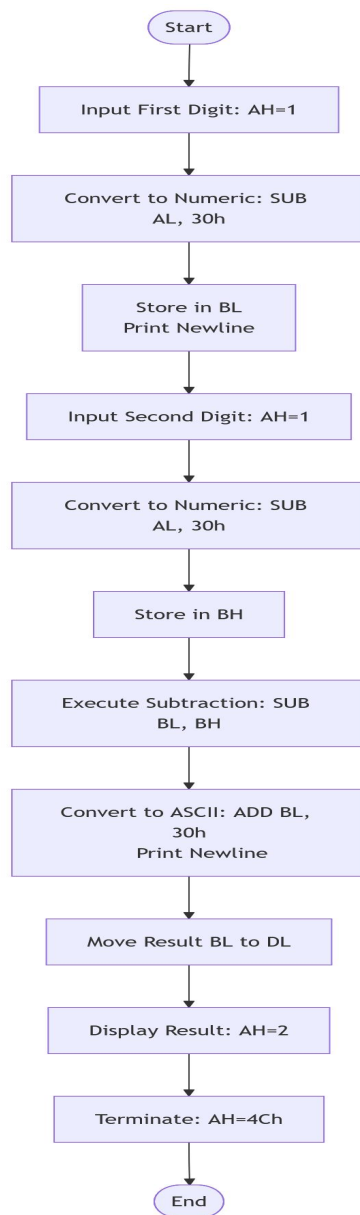
Experiment Name:

Write an Assembly Language program to perform subtraction of two numbers.

Introduction:

Subtraction in the 8086 microprocessor is executed using the **SUB** instruction, which subtracts the source operand from the destination operand. To perform a valid mathematical subtraction, we first convert these ASCII inputs into their decimal equivalents by subtracting **30h** (48 in decimal). After the subtraction is complete, the numerical result must be converted back into an ASCII character by adding **30h** so it can be correctly displayed on the console. This experiment reinforces the understanding of register-to-register operations and data normalization in low-level programming.

Flowchart:



Code:

; an Assembly Language program to perform subtraction of two numbers

ORG 100h

.code

main proc

;first input

MOV AH, 1

INT 21h

SUB AL, 30h

MOV BL, AL

;newline

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;second input

MOV AH, 1

INT 21h

SUB AL, 30h

MOV BH, AL

;Subtraction

SUB BL, BH

ADD BL, 30h

;newline

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;result

MOV DL, BL

MOV AH, 2

INT 21h

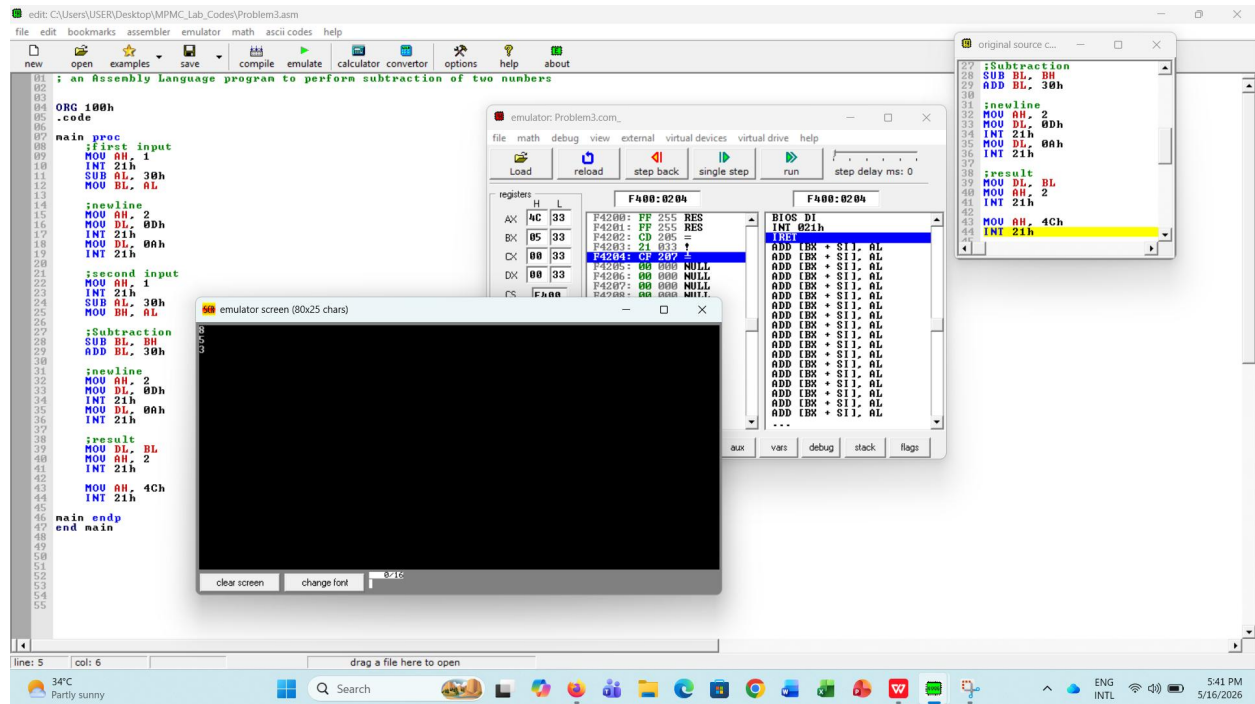
MOV AH, 4Ch

INT 21h

main endp

end main

Output:



Conclusion:

This experiment successfully demonstrated the logic required to perform subtraction in assembly language. By converting ASCII inputs into raw numbers and then back into ASCII for output, we observed how the processor handles data representation. We also practiced using the BL and BH registers to manage multiple pieces of data simultaneously. This logic serves as a foundation for more complex mathematical operations and demonstrates the importance of the SUB and ADD instructions in data transformation.

Experiment No: 04

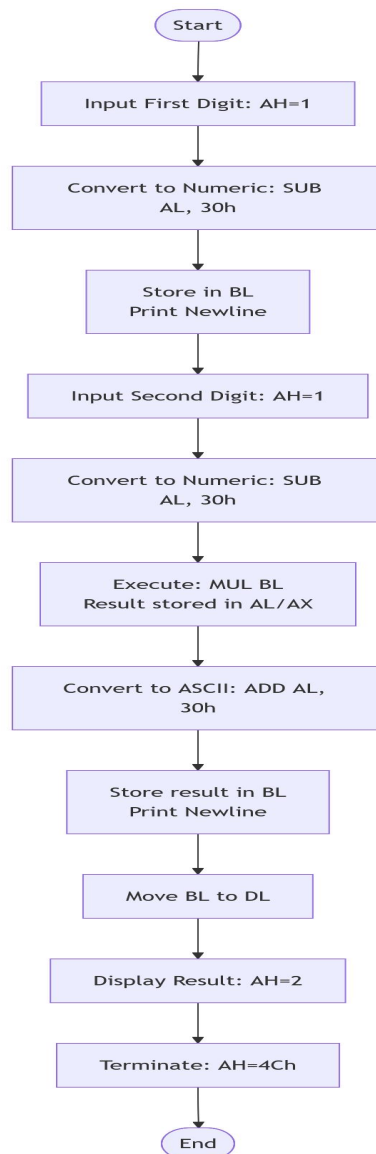
Experiment Name:

Write an Assembly Language program to perform multiplication of two numbers.

Introduction:

Multiplication in the 8086 microprocessor is handled by the **MUL** instruction for unsigned operands. A unique characteristic of this instruction is its implicit use of the Accumulator register. For 8-bit multiplication, the processor automatically multiplies the value in AL by the specified source operand. Since user inputs are captured as ASCII values, they are first normalized by subtracting 30h to obtain the actual numerical digits. The product is then converted back to ASCII by adding 30h for console display. This experiment demonstrates the specific register constraints of the MUL instruction and the process of handling arithmetic results within the 8086 architecture.

Flowchart:



Code:

; an Assembly Language program to perform multiplication of two numbers

ORG 100h

.CODE

main proc

;First Input

MOV AH, 1

INT 21h

SUB AL, 30h

MOV BL, AL

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Second Input

MOV AH, 1

INT 21h

SUB AL, 30h

;Multiplication

MUL BL

ADD AL, 30h

MOV BL, AL

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;result

MOV DL, BL

MOV AH, 2

INT 21h

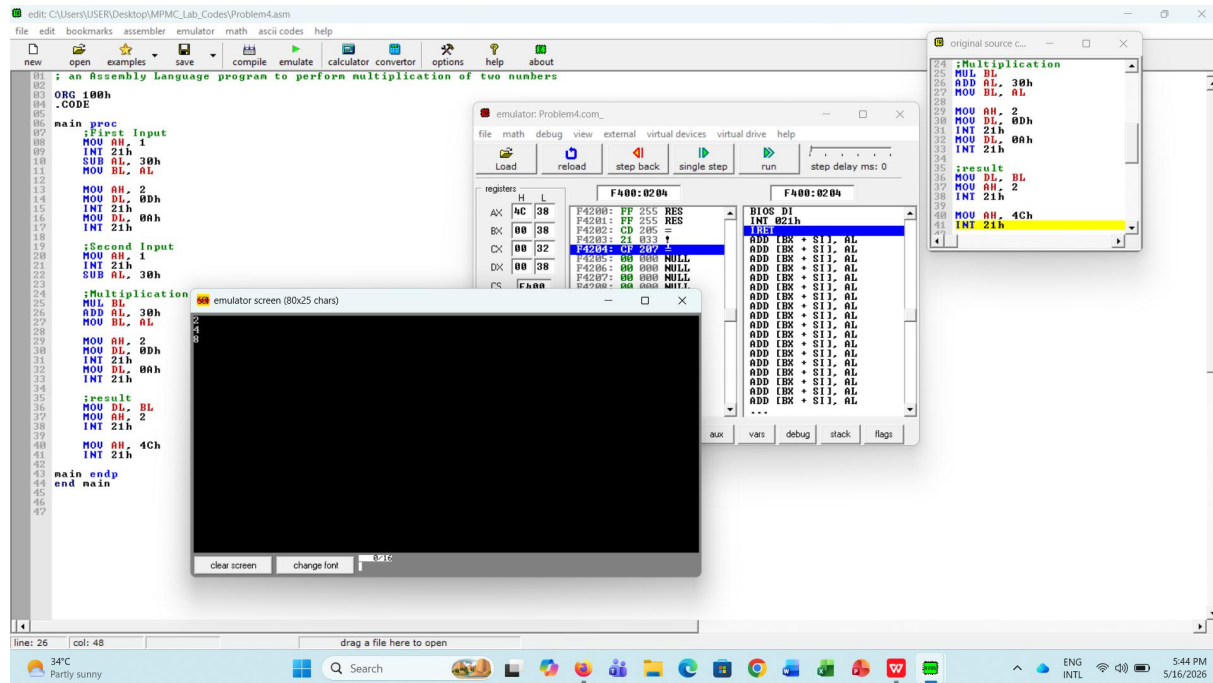
MOV AH, 4Ch

INT 21h

main endp

end main

Output:



Conclusion:

In this experiment, the multiplication of two single-digit numbers is performed successfully. We observed that the MUL instruction implicitly targets the AL register, requiring careful management of data before execution. The experiment also reinforced the necessity of ASCII-to-numeric conversion for arithmetic accuracy. This logic is fundamental for understanding how microprocessors handle more complex mathematical computations and multi-register results.

Experiment No: 05

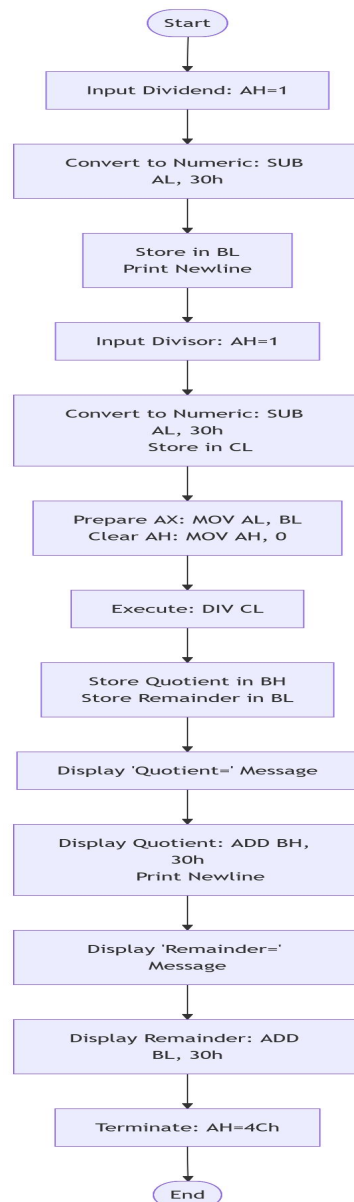
Experiment Name:

Write an Assembly Language program to perform division of two numbers.

Introduction:

Division in the 8086 microprocessor is performed using the **DIV** instruction for unsigned data. Unlike addition or subtraction, the DIV instruction produces two distinct results simultaneously: the **Quotient** and the **Remainder**. When dividing an 8-bit number, the dividend must be placed in the AX register. The instruction then divides AX by the specified divisor, storing the quotient in the **AL** register and the remainder in the **AH** register. In this experiment, user inputs are converted from ASCII to numeric form by subtracting 30h. After the division, both the quotient and remainder are converted back to ASCII (by adding 30h) to be displayed as readable characters on the console.

Flowchart:



Code:

; an Assembly Language program to perform division of two numbers

ORG 100h

.code

main proc

; 1. Input first digit

MOV AH, 1

INT 21h

SUB AL, 30h

MOV BL, AL

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

; 2. Input second digit

MOV AH, 1

INT 21h

SUB AL, 30h

MOV CL, AL

;Division

MOV AL, BL

MOV AH, 0

DIV CL

MOV BH, AL

MOV BL, AH

;Print quotient

MOV AH, 9

MOV DX, OFFSET MSG_Q

INT 21h

MOV DL, BH

ADD DL, 30h

MOV AH, 2

INT 21h

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Print Remainder

MOV AH, 9

MOV DX, OFFSET MSG_R

INT 21h

MOV DL, BL

ADD DL, 30h

MOV AH, 2

INT 21h

MOV AH, 4Ch

INT 21h

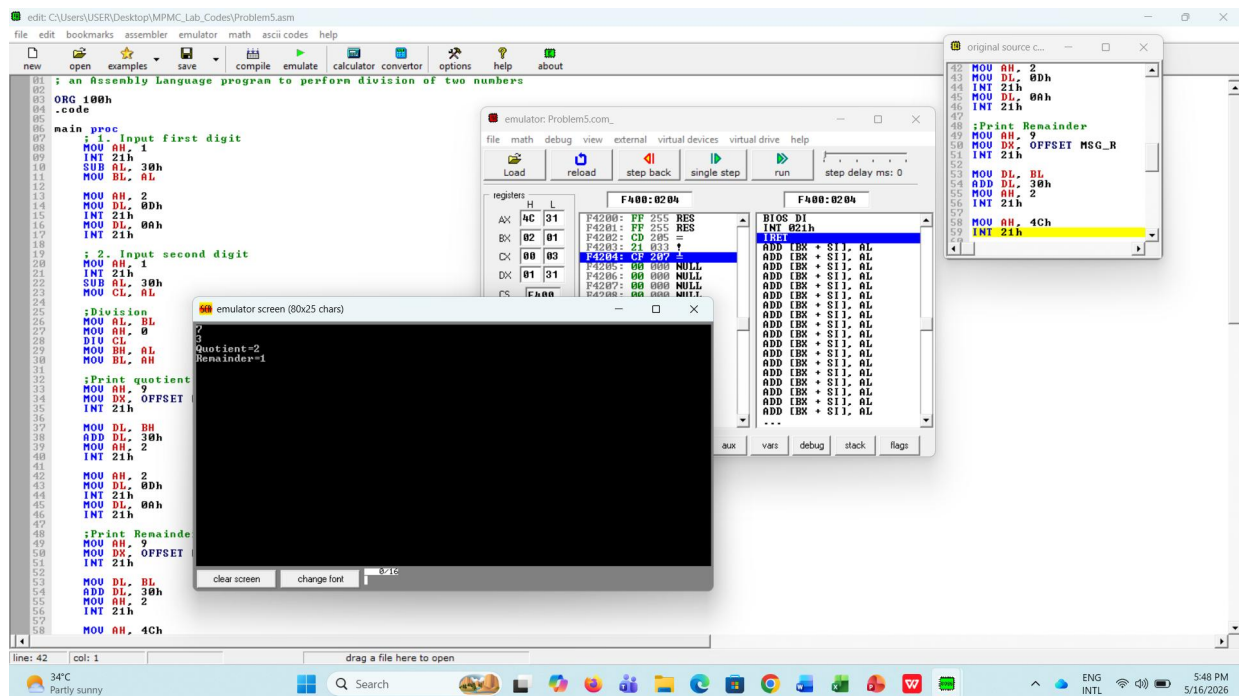
MSG_Q DB 0Dh, 0Ah, 'Quotient=\$'

MSG_R DB 'Remainder=\$'

main endp

end main

Output:



Conclusion:

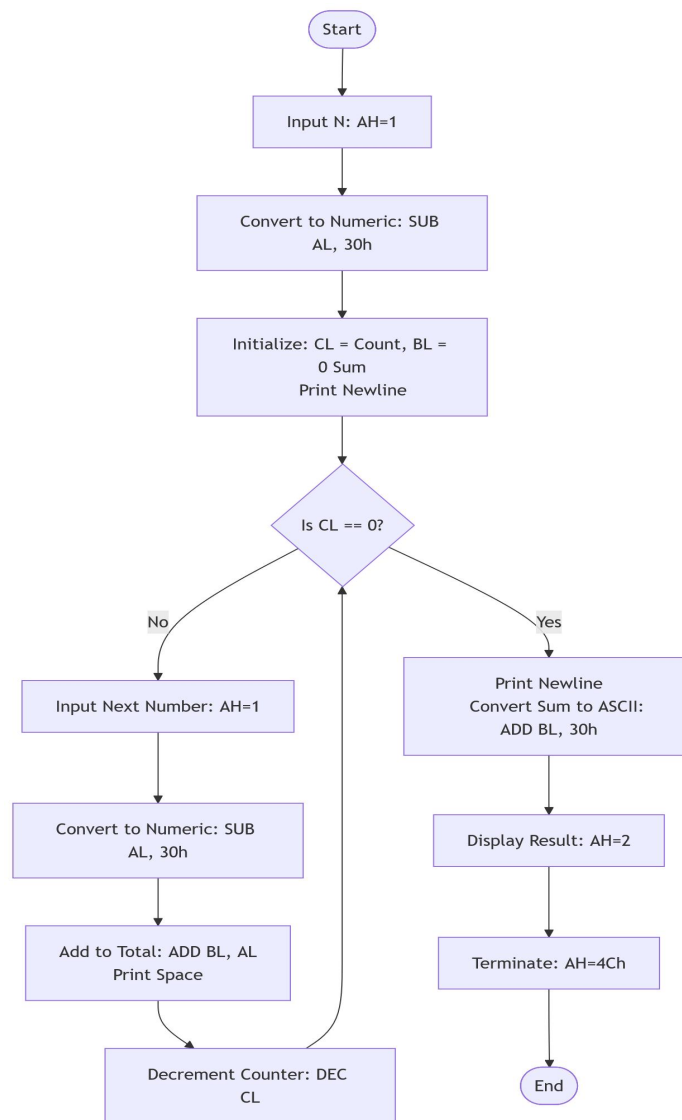
This experiment successfully demonstrated how the 8086 microprocessor handles division operations. The results confirmed that the processor effectively splits the output into two parts: AL for the quotient and AH for the remainder. This experiment proved valuable insight into multi-output instructions and the management of string-based labels using INT 21h function 09h.

Experiment No: 06**Experiment Name:**

Write an Assembly Language program to calculate the sum of N numbers using a loop.

Introduction:

This experiment demonstrates the implementation of iterative logic in the 8086 microprocessor using jump and comparison instructions. To calculate the sum of N numbers, the program first captures the value of N from the user and stores it in the CL register, which acts as a counter. A loop is then established using a label and the **CMP** (Compare) and **JE** (Jump if Equal) instructions to check if the counter has reached zero. Within each iteration, the program takes a single-digit input, converts it from ASCII to numeric form and adds it to a running total stored in the BL register. This process highlights how conditional branching and register manipulation are used to perform repetitive arithmetic tasks in a low-level environment.

Flowchart:**Code:**

;an Assembly Language program to calculate the sum of N numbers using a loop

```
ORG 100h
```

```
.code
```

```
main proc
```

```
    ;print (N)
```

```
    MOV AH, 1
```

```
    INT 21h
```

```
    SUB AL, 30h
```

```
MOV CL, AL
MOV BL, 0
```

```
MOV AH, 2
MOV DL, 0Dh
INT 21h
MOV DL, 0Ah
INT 21h
```

```
;Loop
```

```
L1:
  CMP CL, 0
  JE Result
```

```
;Input numbers
MOV AH, 1
INT 21h
SUB AL, 30h
ADD BL, AL
```

```
; Print a space
MOV AH, 2
MOV DL, 32
INT 21h
```

```
DEC CL
JMP L1
```

```
Result:
```

```
MOV AH, 2
MOV DL, 0Dh
INT 21h
MOV DL, 0Ah
INT 21h
```

```
;Sum
MOV DL, BL
ADD DL, 30h
MOV AH, 2
INT 21h
```

```
MOV AH, 4Ch
INT 21h
```

```
main endp
end main
```

Output:

The screenshot displays an assembly language editor window titled "edit: C:\Users\USER\Desktop\MPMC_Lab_Codes\Problem6.asm". The code is as follows:

```

;ian Assembly Language program to calculate the sun of N numbers using a loop
ORG 100h
.code
main proc
    i:print (N)
    MOV AH, 1
    INT 21h
    SUB AL, 30h
    MOV CL, AL
    MOV BL, 0
    MOV AH, 2
    MOV DL, 00h
    INT 21h
    MOV DL, 00h
    INT 21h
    LI: ;Loop
        CMP CL, 0
        JE Result
    ;input numbers
    MOV AH, 1
    INT 21h
    SUB AL, 30h
    ADD BL, AL
    ; Print a space
    MOV AH, 2
    MOV DL, 32
    INT 21h
    DEC CL
    JMP LI
Result:
    MOV AH, 2
    MOV DL, 00h
    INT 21h
    MOV DL, 00h
    INT 21h
    ;Sum
    MOV DL, BL
    ADD DL, 30h
    MOV AH, 2
    INT 21h
    MOV AH, 4Ch
    INT 21h
main endp
end main

```

The emulator window shows the execution of the program. The registers window displays the following values:

Register	Value
AX	4C 37
BX	00 07
CX	00 00
DX	00 37
CS	F400

The BIOS window shows the following values:

BIOS DI	Value
INT 021h	...

The emulator screen (80x25 chars) displays the output of the program:

```

0 0 1 3 1

```

Conclusion:

In this experiment, a loop to perform cumulative addition is implemented successfully. By utilizing the DEC and CMP instructions, we controlled the execution flow based on a user-defined count. The experiment shows the importance of initializing the sum register (BL) to zero and demonstrated how a single register can be used to maintain a running total. This logic is fundamental for processing arrays, data streams and performing complex mathematical summations in assembly language.

Experiment No: 07

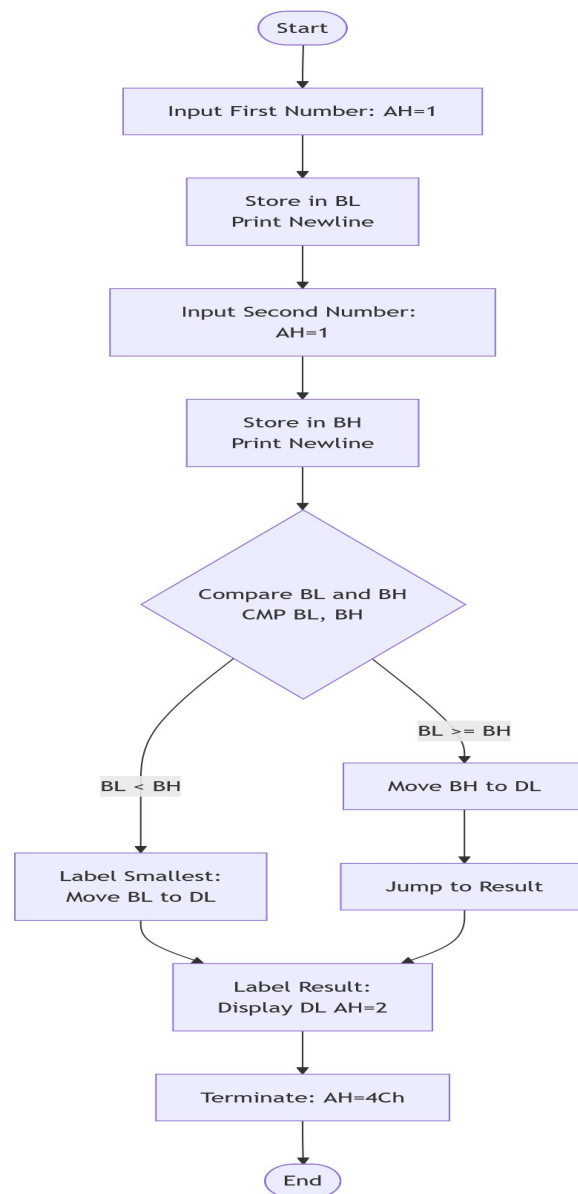
Experiment Name:

Write an Assembly Language program to find the smallest number between two given numbers.

Introduction:

Selection and decision-making in the 8086 microprocessor are achieved through the combination of the **CMP** (compare) instruction and conditional jump instructions. The **CMP** instruction performs a subtraction between two operands to update the processor's status flags (such as the Zero Flag and Sign Flag) without modifying the actual values in the registers. This experiment uses the **JL** (Jump if Less) instruction, which checks these flags to determine the execution path. By comparing two user-provided characters, the program branches to different labels to isolate the smaller value and move it into the DL register for output. This logic is fundamental for implementing algorithms that require sorting or range checking.

Flowchart:



Code:

;an Assembly Language program to find the smallest number between two given numbers

ORG 100h

.code

main proc

;Input first number

MOV AH, 1

INT 21h

MOV BL, AL

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Input second number

MOV AH, 1

INT 21h

MOV BH, AL

MOV AH, 2

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Compare numbers

CMP BL, BH

JL smallest

MOV DL, BH

JMP result

smallest:

MOV DL, BL

result:

MOV AH, 2

INT 21h

MOV AH, 4Ch

INT 21h

main endp

end main

Output:

The screenshot displays an x86 assembly emulator with the following components:

- Source Code (Problem7.asm):**

```

01 ;an Assembly Language program to find the smallest number between two given numbers
02
03
04
05 .code
06
07 main proc
08     ;input first number
09     MOV AH, 1
10     INT 21h
11     MOV BL, AL
12
13     MOV AH, 2
14     MOV DL, 00h
15     INT 21h
16     MOV DL, 00h
17     INT 21h
18
19     ;input second number
20     MOV AH, 1
21     INT 21h
22     MOV BH, AL
23
24     MOV AH, 2
25     MOV DL, 00h
26     INT 21h
27     MOV DL, 00h
28     INT 21h
29
30     ;Compare numbers
31     CMP BL, BH
32     JL smallest
33
34     MOV DL, BH
35     JMP result
36
37 smallest:
38     MOV DL, BL
39
40 result:
41     MOV AH, 2
42     INT 21h
43     MOV AH, 4Ch
44     INT 21h
45
46 main endp
47 end main

```
- Registers:**

Register	Value
AX	37
CX	37
DX	37
- Memory:**

Address	Value
F400:0204	FF 255
F400:0205	FF 255
F400:0206	CB 285
F400:0207	21 033
F400:0208	00 000
F400:0209	00 000
F400:020A	00 000
F400:020B	00 000
F400:020C	00 000
F400:020D	00 000
F400:020E	00 000
F400:020F	00 000
F400:0210	00 000
F400:0211	00 000
F400:0212	00 000
F400:0213	00 000
F400:0214	00 000
F400:0215	00 000
F400:0216	00 000
F400:0217	00 000
F400:0218	00 000
F400:0219	00 000
F400:021A	00 000
F400:021B	00 000
F400:021C	00 000
F400:021D	00 000
F400:021E	00 000
F400:021F	00 000
- Output Window:**

```

28 INT 21h
29
30 ;Compare numbers
31 CMP BL, BH
32 JL smallest
33
34 MOV DL, BH
35 JMP result
36
37 smallest:
38 MOV DL, BL
39
40 result:
41 MOV AH, 2
42 INT 21h
43 MOV AH, 4Ch
44 INT 21h
45

```

Conclusion:

In this experiment, we successfully implemented a decision-making algorithm to identify the smaller of two inputs. We observed that the CMP instruction effectively prepares the processors for branching, while conditional jumps like JL allow the program to skip specific blocks of code based on numerical relationships. This experiment reinforced the concept of “Labels” and “unconditional jumps” (JMP) to maintain a clean execution flow after a decision is made. Understanding these control structures is essential for developing logic-driven assembly programs.

Experiment No: 08

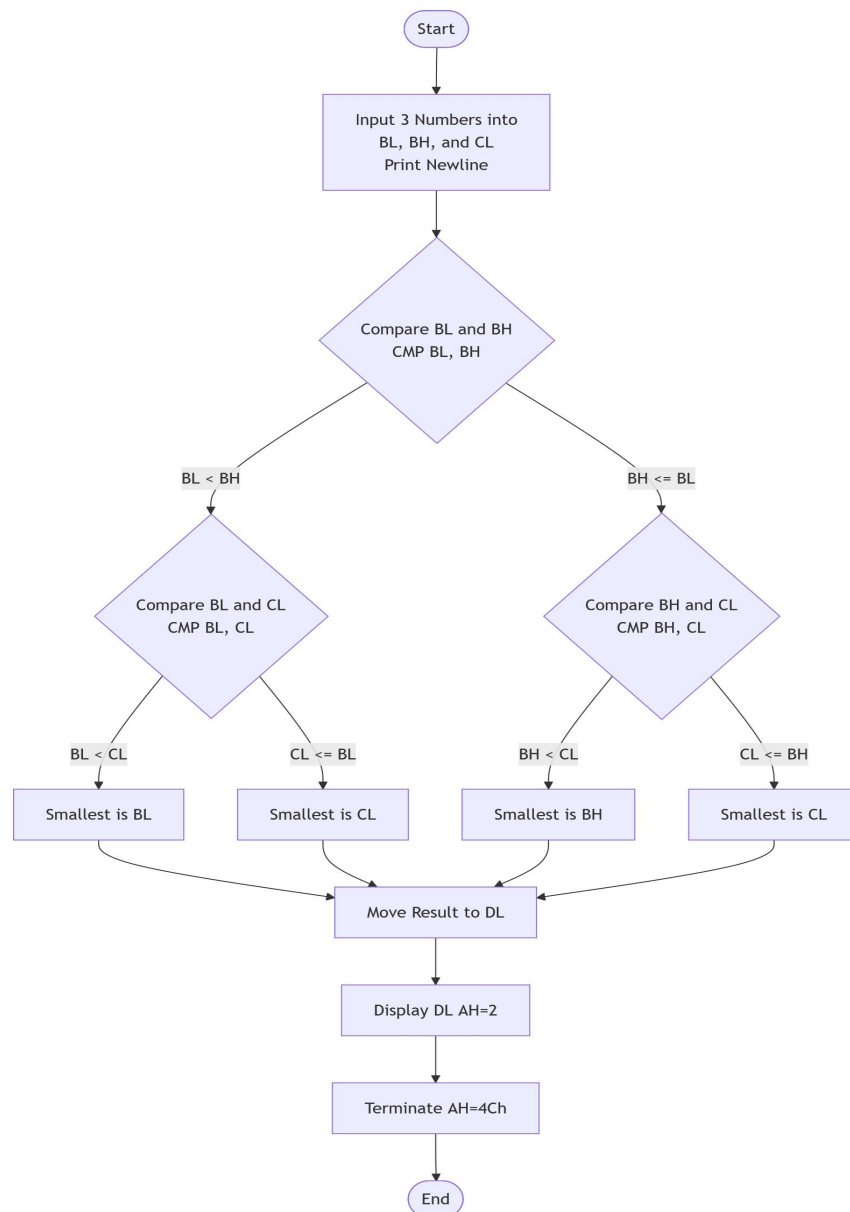
Experiment Name:

Write an Assembly Language program to find the smallest number among three given numbers.

Introduction:

Finding the minimum value among multiple variables is a common task in data processing. In 8086 Assembly, this is achieved through nested conditional logic using the **CMP** (compare) instruction and conditional jumps like **JL** (Jump if Less). The process is first comparing two numbers to find a temporary minimum, and then comparing that result with the third number. This experiment demonstrates how to manage multiple registers (**BL**, **BH** and **CL**) and how to structure code with multiple labels and unconditional jumps to ensure that only the final smallest value is moved to the **DL** register for output.

Flowchart:



Code:

;an Assembly Language program to find the smallest number among three given numbers

ORG 100h

.code

main proc

;Input first number

MOV AH, 1

INT 21h

MOV BL, AL

MOV AH, 2 ;space

MOV DL, 32

INT 21h

;Input second number

MOV AH, 1

INT 21h

MOV BH, AL

MOV AH, 2 ;space

MOV DL, 32

INT 21h

;Input third number

MOV AH, 1

INT 21h

MOV CL, AL

MOV AH, 2 ;newline

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Compare BL and BH

CMP BL, BH

JL BL_CL ;If BL is smaller than BH, checking with CL

;Else, BH is smaller than BL, checking BH with CL

CMP BH, CL

JL BH_Smallest ;If BH < CL, then BH is the smallest

MOV DL, CL ;Else, CL is the smallest

JMP Result

BL_CL:

CMP BL, CL

JL BL_Smallest ;If BL < CL, then BL is the smallest
 MOV DL, CL ;Else, CL is the smallest
 JMP Result

BH_Smallest:
 MOV DL, BH
 JMP Result

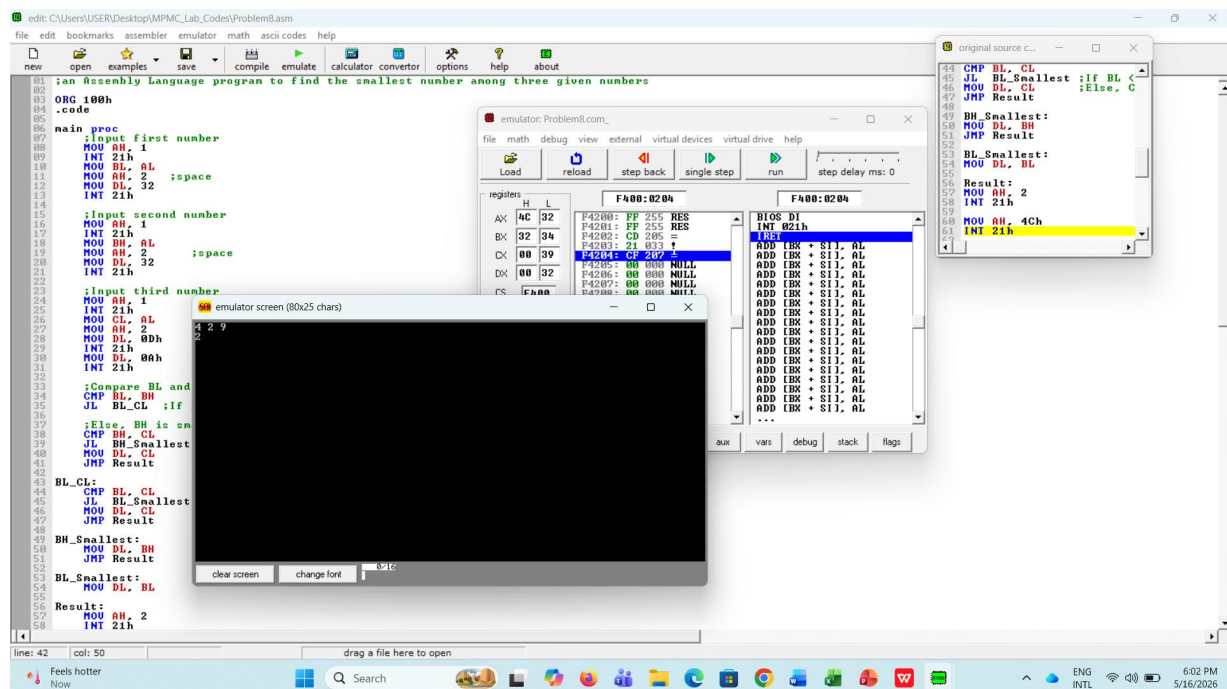
BL_Smallest:
 MOV DL, BL

Result:
 MOV AH, 2
 INT 21h

 MOV AH, 4Ch
 INT 21h

main endp
 end main

Output:



Conclusion:

This experiment successfully implemented the logic to determine the smallest value among three inputs. By using a series of comparisons and branches, we observed how the processor can evaluate complex conditions using simple 8-bit registers. The use of multiple labels (**BL_CL**, **BH_Smallest**, **BL_Smallest**) highlighted the importance of clear program flow and the role of the JMP instruction in preventing the execution of unintended code blocks. This exercise is critical for understanding algorithmic decision-making at the machine level.

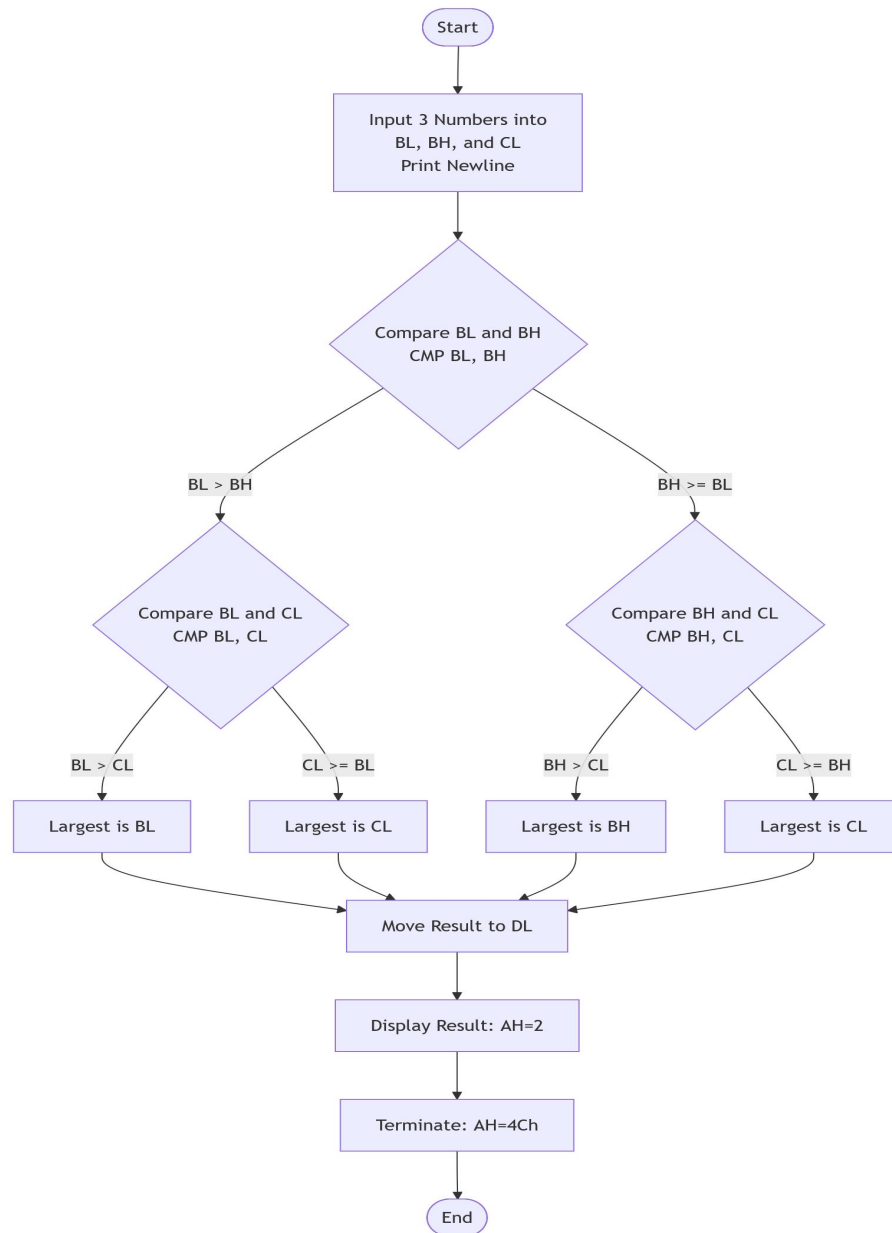
Experiment No: 09

Experiment Name:

Write an Assembly Language program to find the largest number among three given numbers.

Introduction:

Determining the maximum value from a set of inputs is a fundamental logic operation in assembly language. This experiment utilizes the 8086 microprocessor's **CMP** (Compare) instruction in conjunction with the **JG** (Jump if Greater) conditional jump instruction. The logic follows a comparison-based elimination strategy; first, two registers (BL and BH) are compared to identify the potential maximum. The winner of that comparison is then compared against the third register (CL). By using specific labels like **BL_CL**, **BH_Largest** and **BL_Largest**, the program successfully branches to the correct path to isolate the highest value. This approach demonstrates how complex "if-else" structures from high-level languages are translated into a series of status flag checks and jumps at the hardware level.

Flowchart:

Code:

;an Assembly Language program to find the largest number among three given numbers

ORG 100h

.code

main proc

;Input first number

MOV AH, 1

INT 21h

MOV BL, AL

MOV AH, 2 ;space

MOV DL, 32

INT 21h

;Input second number

MOV AH, 1

INT 21h

MOV BH, AL

MOV AH, 2 ;space

MOV DL, 32

INT 21h

;Input third number

MOV AH, 1

INT 21h

MOV CL, AL

MOV AH, 2 ;newline

MOV DL, 0Dh

INT 21h

MOV DL, 0Ah

INT 21h

;Compare BL and BH

CMP BL, BH

JG BL_CL ;If BL is greater than BH, check BL with CL

;Else, BH is greater than or equal to BL, check BH with CL

CMP BH, CL

JG BH_Largest ;If BH > CL, then BH is the largest

MOV DL, CL ;Else, CL is the largest

JMP RESULT

BL_CL:

CMP BL, CL

JG BL_Largest ;If BL > CL, then BL is the largest

Conclusion:

Through this experiment, we successfully implemented a selection algorithm to identify the largest among three numerical characters. We learned how to utilize the **JG** instruction, which depends on the state of the overflow, sign and zero flags after a comparison. The logic confirmed that regardless of the order of input, the program effectively narrows down the largest value through sequential branching. This experiment further solidified our understanding of how to use multiple registers and jump instructions to control the execution flow of a microprocessor.

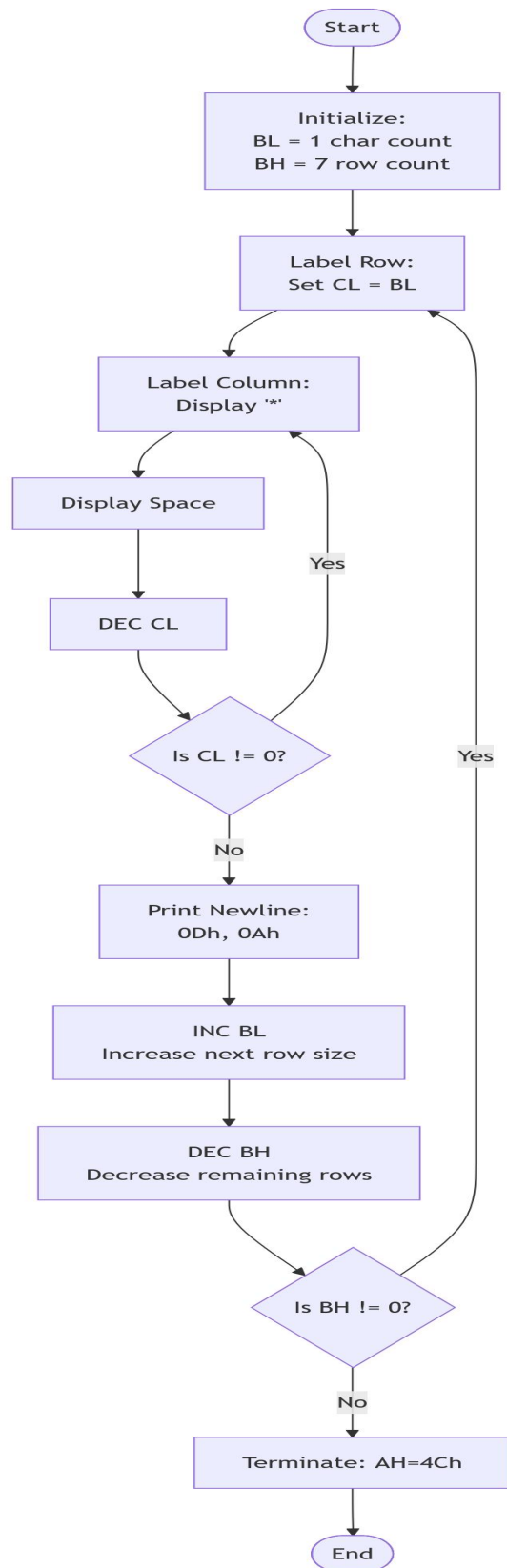
Experiment No: 10

Experiment Name:

Write an Assembly Language program to print a simple pattern (e.g., triangle, square, or pyramid). [**Here, Printing a Triangle using (*)**]

Introduction:

Generating geometric patterns on the console is a classic way to demonstrate the use of nested loops and control flow in assembly language. In this experiment, a right-angled triangle pattern is printed using '*' characters. The program utilizes two loops; an **outer loop** (Row) that manages the number of lines to be printed and an **inner loop** (Column) that determines the number of characters to display on each specific line. By using the **INC** (Increment) instruction on the inner loop limit and the **DEC** (Decrement) instruction on the row counter, the program creates a dynamic structure where each subsequent row contains one more character than the previous one.

Flowchart:

Code:

;an Assembly Language program to print a simple pattern (e.g., triangle, square, or pyramid)
 ; Here i'm creating a Triangle with (*):

```
ORG 100h
```

```
.code
```

```
main proc
```

```
    MOV BL, 1
```

```
    MOV BH, 7 ;numbers of row
```

```
Row:
```

```
    MOV CL, BL
```

```
Column:
```

```
    MOV AH, 2
```

```
    MOV DL, '*'
```

```
    INT 21h
```

```
    MOV DL, 32
```

```
    INT 21h
```

```
    DEC CL
```

```
    JNZ Column
```

```
    MOV AH, 2
```

```
    MOV DL, 0Dh
```

```
    INT 21h
```

```
    MOV DL, 0Ah
```

```
    INT 21h
```

```
    INC BL
```

```
    DEC BH
```

```
    JNZ Row
```

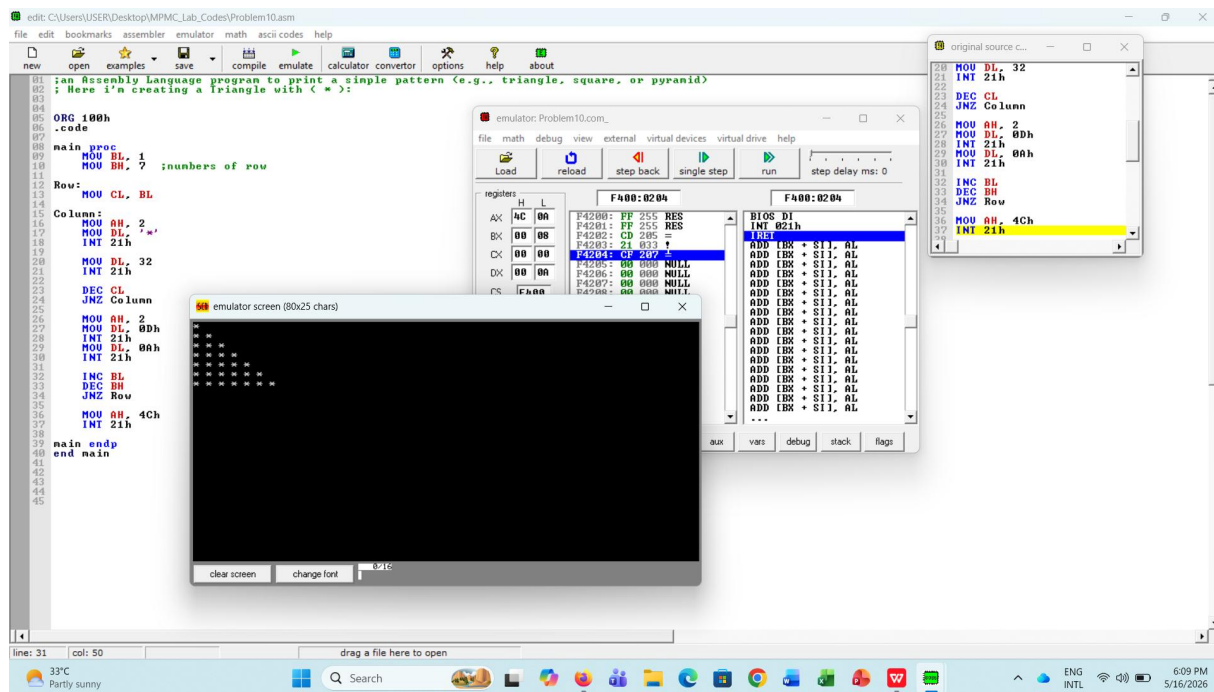
```
    MOV AH, 4Ch
```

```
    INT 21h
```

```
main endp
```

```
end main
```

Output:



Conclusion:

This experiment successfully demonstrates the implementation of nested loops in 8086 Assembly. We learned how to coordinate two different registers (BL and BH) to manage rows and columns independently. The use of conditional jumps like JNZ (Jump if not Zero) allowed us to repeat character printing until the desired width was achieved for each row. This logic is essential for understanding how microprocessors handle 2D data structures and complex console formatting tasks.

Experiment No: 11

Experiment Name:

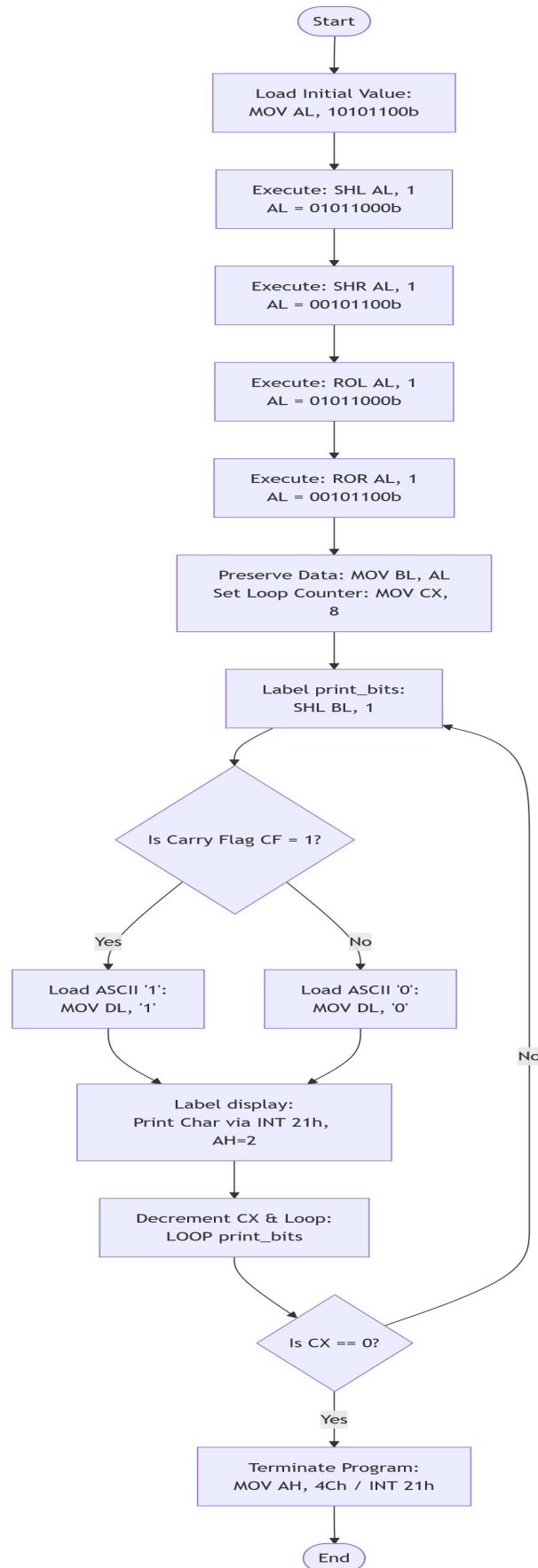
Write a program to implement the following instructions: SHL, SHR, ROL & ROR.

Introduction:

Bit-wise Operations manipulate individual bits within a processor register. Here:

- **SHL (Shift Left):** Shifts all bits to the left by the specified count. The highest bit enters the Carry Flag (CF) and a (0) fills the vacated rightmost position.
- **SHR (Shift Right):** Shifts all bits to the right. The lowest bit enters the Carry Flag (CF), and a (0) fills the vacated leftmost position.
- **ROL (Rotate Left):** Rotates bits to the left. The highest bit wraps around to fill the vacated rightmost position and copies into the Carry Flag (CF).
- **ROR (Rotate Right):** Rotates bits to the right. The lowest bit wraps around to fill the vacated leftmost position and copies into the Carry Flag (CF).

This program demonstrates these instructions on an 8-bit binary value and extracts the final bit configuration using a loop to display it as ASCII text on the console.

Flowchart:

Code:

; a program to implement the following instructions : SHL, SHR, ROL, and ROR

```

ORG 100h
.CODE

main proc
    MOV AL, 10101100b ; Initial value
    SHL AL, 1          ; Shift Left (AL becomes 01011000b)
    SHR AL, 1          ; Shift Right (AL becomes 00101100b)
    ROL AL, 1          ; Rotate Left (AL becomes 01011000b)
    ROR AL, 1          ; Rotate Right (AL becomes 00101100b)

    MOV BL, AL
    MOV CX, 8

print_bits:
    SHL BL, 1
    JC print_one

    MOV DL, '0'
    JMP display

print_one:
    MOV DL, '1'

display:
    MOV AH, 2
    INT 21h
    LOOP print_bits

    MOV AH, 4Ch
    INT 21h
main endp
end main

```

Output:

The screenshot displays an 8086 emulator interface. The main window shows assembly code for a program that implements SHL, SHR, ROL, and ROR instructions. The code includes comments explaining the bit-shifting and rotation operations. A separate window shows the original source code. The registers window displays the current state of the registers, including the AL register. The terminal window shows the output of the program, displaying the binary representation of the initial value and the results of the bit-shifting and rotation operations.

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